

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I M. Varshith Reddy(192524174)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Varshith Reddy

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 5	/ 6	/ 4	/ 3	/ 2	
2	/ 5	/ 6	/ 4	/ 3	/ 2	
3	/ 5	/ 6	/ 4	/ 3	/ 2	
4	/ 5	/ 6	/ 4	/ 3	/ 2	
5	/ 5	/ 6	/ 4	/ 3	/ 2	
OVERALL RUBRICS VALUE						
	/ 5	/ 5	/ 4	/ 4	/ 2	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

Deduce the most likely original message.

Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality. Demonstrate how the encryption process transforms the message step-by-step.

Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

Analyze how investigators could detect the presence of hidden data within the image.

Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
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Below are **complete answers** for all **5 application-based questions** from your **CSA5113 – Cryptography and Network Security Assessment Tool 1**.

Q1. Event Communication Security (Caesar Cipher – Shift of 4)

Given

Plaintext: **MEET AT NOON**

Shift = **+4**

Encryption

Plaintext M E E T A T N O O N

Shift +4 **Q I I X E X R S S R**

Ciphertext

QIIX EX RSSR

Decryption

An authorized receiver knows the shift value is **4**.

Subtract **4** from each encrypted letter.

Ciphertext Q I I X E X R S S R

Shift **-4** **M E E T A T N O O N**

Recovered message:

MEET AT NOON

Result

- Encryption method: Caesar Cipher
 - Ciphertext: **QIIX EX RSSR**
 - Original message recovered successfully using a shift of **-4**.
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Q2. Intercepted Suspicious Message (Atbash Cipher)

Ciphertext

GSRH RH Z HVXIVG

The message follows the **Atbash Cipher**, where letters are reversed:

A↔Z

B↔Y

C↔X

...

Decryption

Cipher	G		S		R		H			R		H			Z			H		V		X		I		V		G	
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---
Plain	T		H		I		S			I		S			A			S		E		C		R		E		T	

Original Message

THIS IS A SECRET

Justification

- The repeated pattern **RH** becomes **IS**.
- The last word **HVXIVG** becomes **SECRET**.
- The Atbash substitution exactly converts the ciphertext into meaningful English.

Result

Recovered plaintext:

THIS IS A SECRET

Q3. Secure Messaging System (Vigenère Cipher)

Plaintext

HELLO

Keyword

KEY

Repeat keyword:

Plaintext : H E L L O

Keyword : K E Y K E

Letter Values

Letter Value

H	7
E	4
L	11
L	11
O	14

Keyword

Letter Value

K	10
E	4
Y	24
K	10
E	4

Encryption

Formula:

$$\text{Cipher} = (\text{Plain} + \text{Key}) \bmod 26$$

Plain Key Calculation Cipher

H	K	$7+10=17$	R
E	E	$4+4=8$	I
L	Y	$11+24=35 \rightarrow 9$	J
L	K	$11+10=21$	V
O	E	$14+4=18$	S

Ciphertext

RIJVS

Keyword Effect

- The keyword repeats until the message length is reached.
- Every plaintext letter is shifted by a different amount.

- This makes Vigenère Cipher stronger than Caesar Cipher because identical plaintext letters may encrypt differently.

Result

Plaintext: **HELLO**

Keyword: **KEY**

Ciphertext: **RIJVS**

Q4. Suspicious Digital Asset Investigation (Steganography)

How investigators detect hidden data

1. LSB Analysis

- Examine the Least Significant Bits of image pixels.
- Unexpected changes may indicate hidden information.

2. Statistical Analysis

- Compare pixel distribution with normal images.
- Irregular patterns suggest embedded data.

3. Steganalysis Tools

- Use software such as StegExpose or StegDetect to scan images.

4. Metadata Inspection

- Check image properties for unusual metadata or hidden attachments.

Two techniques to strengthen hidden communication

Technique 1 – Encrypt before embedding

- Encrypt the message using AES or another strong algorithm before hiding it.
- Even if extracted, the content remains unreadable.

Technique 2 – Randomized embedding

- Use a secret key to choose random pixel locations for embedding.
- This reduces the chance of detection by steganalysis tools.

Result

Combining **encryption** with **randomized steganography** greatly improves the confidentiality and security of hidden communications.

Q5. Legacy Encryption in Corporate Communication (Rail Fence Cipher)

Given Ciphertext

WECRLTEERDSOEFEAOCAIVDEN

(a) Original Message

This is a **3-Rail Fence Cipher**.

Writing the ciphertext back into three rails and reading row-wise gives:

W . . . E . . . C . . . R

. E . R . D . S . O . E .

.. L . . . T . . . E ..

The recovered plaintext is:

WE ARE DISCOVERED FLEE AT ONCE

(b) Why this method is vulnerable

1. It does not change the letters, only their positions.
 2. Frequency analysis can still be performed.
 3. The number of rails is easy to guess.
 4. Modern computers can quickly brute-force all possible rail values.
 5. It provides very weak security compared with modern encryption algorithms.
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(c) Practical enhancement

A better approach is to:

- Encrypt the message using **AES-256** before transmission.
- Alternatively, combine transposition with substitution to create stronger protection.

AES provides:

- High confidentiality
- Strong resistance to brute-force attacks
- Widely accepted modern security

Result

Recovered plaintext:

WE ARE DISCOVERED FLEE AT ONCE

The Rail Fence Cipher is insecure for modern communication and should be replaced with strong encryption such as **AES**.